TROPICAMIDE

Can precipitate angle-closure

No driving afterwards

CHEMICAL NAME (RS)-N-ethyl-3-hydroxy-2-phenyl-N-(pyridin-4-ylmethyl)propanamide
CLASS Short-acting antimuscarinic mydriatic
TRADE NAMES Mydriacyl · Mydriaticum · Tropicamide Minims · Mydrin-P (with phenylephrine)

THERAPEUTIC USES / INDICATIONS

- Pupil dilation (mydriasis) for fundus examination and retinal imaging.
- Pre-operative mydriasis (cataract surgery, with phenylephrine).

ADMINISTRATION

- **Eye drops:** 1 drop, repeat after 5 min if needed; punctal occlusion (especially in infants).
- Single-use Minims to avoid contamination.

DOSAGES

- 0.5–1%: 1–2 drops 15–20 min before examination.
- Infants: 0.5% (often with phenylephrine 2.5%).

PHARMACOKINETICS

- **Onset:** 15–30 min. **Duration:** 4–8 h (shortest-acting mydriatic).
- Weak cycloplegia (recovery 6 h).

PHARMACODYNAMICS

- Blocks **muscarinic receptors** of iris sphincter (→ mydriasis) and ciliary muscle (→ weak cycloplegia).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Blurred vision, glare/light sensitivity for several hours.
- Transient stinging.
- Dry mouth.

ADVERSE EFFECTS (SERIOUS)

- Acute angle-closure glaucoma (narrow angles); eye pain, red eye, haloes, vomiting.
- Systemic anticholinergic effects in infants (tachycardia, flushing, irritability).
- Psychotic reactions (rare, children).

CONTRAINDICATIONS & CAUTIONS

- Narrow (shallow) anterior chamber angle / angle-closure glaucoma.
- Hypersensitivity.

KEY INTERACTIONS

- **Other anticholinergics:** additive systemic effects.

NURSING MANAGEMENT

- Assess for narrow angles (history of glaucoma attacks) before use.
- Warn about blurred vision; arrange transport.
- Monitor infants for systemic effects.

PATIENT EDUCATION

- Do not drive for at least 4–6 hours (until vision is clear); bring sunglasses.
- Seek urgent help for eye pain, red eye, headache with vomiting or haloes.

KEY POINTS

- Short-acting antimuscarinic mydriatic.
- Risk of angle-closure glaucoma.
- Dilates pupil for fundoscopy; lasts 4–8 h.
- No driving until vision clears.